



WORK SHEET - 29

SOT + TE + AUC + DE

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 In a triangle ABC, $A - B = \frac{2\pi}{3}$ and $R = 8r$ where R is the radius of circumcircle and r is the radius of incircle, then $\cos C$ is equal to
(A) $\frac{7}{8}$ (B) $\frac{1}{6}$ (C) $\frac{1}{16}$ (D) $\frac{5}{8}$
- Q.2 AD, BE, CF are the medians of triangle ABC whose centroid is G. If the points A, F, G, E are concyclic then
(A) $2b^2 = a^2 + c^2$ (B) $2c^2 = a^2 + b^2$ (C) $2a^2 = b^2 + c^2$ (D) $2b^2 = 2a^2 + c^2$
- Q.3 In a triangle ABC, if a, c, A are given $b_1 = 2b_2$ where b_1, b_2 are two values of the third side then the value of $\frac{c\sqrt{1+8\sin^2 A}}{a}$ is equal to
(A) 5 (B) 3 (C) 2 (D) 1
- Q.4 If a, b, c the sides of a triangle ABC (in that order) are in A.P. and $\frac{2!}{1!9!} + \frac{2!}{3!7!} + \frac{1!}{5!5!} = \frac{8^a}{(2b)!}$ then inradius of the triangle ABC is equal to
(A) $\frac{\sqrt{7}}{2}$ (B) $\frac{\sqrt{5}}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$
- Q.5 The number of solutions of $2\sin^3 x + 2\cos^3 x - 3\sin 2x + 2 = 0$ in $[0, 4\pi]$ is
(A) 2 (B) 3 (C) 4 (D) 5
- Q.6 The particular solution of $\cos y \, dx + (1 + 2e^{-x}) \sin y \, dy = 0$ when $x = 0, y = \frac{\pi}{4}$ is
(A) $e^x - 2 = -\sqrt{2} \cos y$ (B) $e^x + 3 = 4\sqrt{2} \cos y$
(C) $e^x + 2 = 3\sqrt{2} \cos y$ (D) $e^x - 3 = -2\sqrt{2} \sin y$
- Q.7 $y \, dx - dy = e^{-x} y^4 \, dy$ and $y = 1$ at $x = -\ln 3$ then at $y = 3, x = \ln k$, k is
(A) 21 (B) 24 (C) 27 (D) 81
- Q.8 The area bounded by $f(x) = \tan^{-1}(\tan x) + \cos^{-1}(\cos x)$ and x -axis when $x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ is equal to
(A) π^2 (B) $\frac{3\pi^2}{4}$ (C) $\frac{\pi^2}{4}$ (D) $2\pi^2$



- Q.9 The area enclosed by the curve $y=0$, $y=f(x)$, $x \geq \frac{-1}{2}$, where $f(x) = \cos^{-1}(4x^3 - 3x)$ is
- (A) $\frac{3\sqrt{3}}{2}$ (B) 1 (C) $\frac{\sqrt{3}}{2}$ (D) $2\sqrt{3}$
- Q.10 If there is no point on the curve $2 \sin y + 8 \sin^3 x = a$ which also lies on the line $y=3x$, then
- (A) $a \in (-\infty, -6) \cup (6, \infty)$ (B) $a \in \{-6, 6\}$
 (C) $a \in (-6, 6)$ (D) $a \in \{-3, 3\}$

[PARAGRAPH TYPE]

Q.11 to Q.13 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 11 to 13

ABC is a triangle in which $a=6$, $b=3$, $\cos(A-B) = \frac{4}{5}$, AD is median through A, $\angle BAD = \theta$.

- Q.11 Triangle ABC is
- (A) isosceles (B) right angled
 (C) obtuse angled (D) isosceles right angled
- Q.12 Length of median AD is equal to
- (A) 3 (B) 6 (C) $3\sqrt{2}$ (D) $3\sqrt{5}$
- Q.13 $\sin \theta$ is equal to
- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{\sqrt{5}}$ (D) $\frac{1}{\sqrt{10}}$

[INTEGER TYPE]

Q.1 to Q.6 are "Integer Type" questions. (The answer to each of the questions are upto **4 digits**) [$6 \times 5 = 30$]

- Q.14 Let $f: [0, 1] \rightarrow \left[0, \frac{1}{2}\right]$ be a function such that $f(x)$ is a polynomial of 2^{nd} degree satisfy the following conditions $f(0)=0$, and has a maximum value of $\frac{1}{2}$ at $x=1$, also $f\left(\frac{1}{2}\right) = \frac{3}{8}$. If A be the area bounded by $y=f(x)$, $y=f^{-1}(x)$ and the line $2x+2y-3=0$ in Ist quadrant then find the value of $24A$.
- Q.15 Find the number of ordered pair (x, y) satisfy $\sin x + \sin y = \sin(x+y)$ and $|x| + |y| = 1$.
- Q.16 Find the number of solution of trigonometric equation $\tan 4x \cos 7x = \sin 7|x|$ in $[-\pi, \pi]$.
- Q.17 If the area of the region bounded by the curve $y=x^2$, $y=\sec^{-1}[-\sin^2 x]$ is $\frac{4\pi\sqrt{\pi}}{k}$, then find the value of k .
- [Note : $[x]$ denotes the greatest integer less than or equal to x .]

Q.18 If solution of equation $\frac{dy}{dx} = y + \int_0^2 y dx$ is $y(x)$, then find the value $[|y(2)|]$.
 [Note : Given $y(0) = 1$ and $[x]$ denotes greatest integer less than or equal to x .]

Q.19 If the area bounded by circle $x^2 + y^2 = 4$, the parabola $y = x^2 + x + 1$ and the curve $y = \left[\sin^2 \frac{x}{4} + \cos \frac{x}{4} \right]$ and x-axis is $\left(\sqrt{3} + \frac{2\pi}{3} - \frac{1}{k} \right)$ then find the value of k .
 [Note : $[x]$ denotes the greatest integer less than or equal to x .]

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ANSWER KEY

Q.1	A	Q.2	C	Q.3	B	Q.4	C	Q.5	C	Q.6	C	Q.7	C
Q.8	B	Q.9	A	Q.10	A	Q.11	B	Q.12	C	Q.13	D	Q.14	5
Q.15	0006	Q.16	0015	Q.17	0003	Q.18	0004	Q.19	0006				

